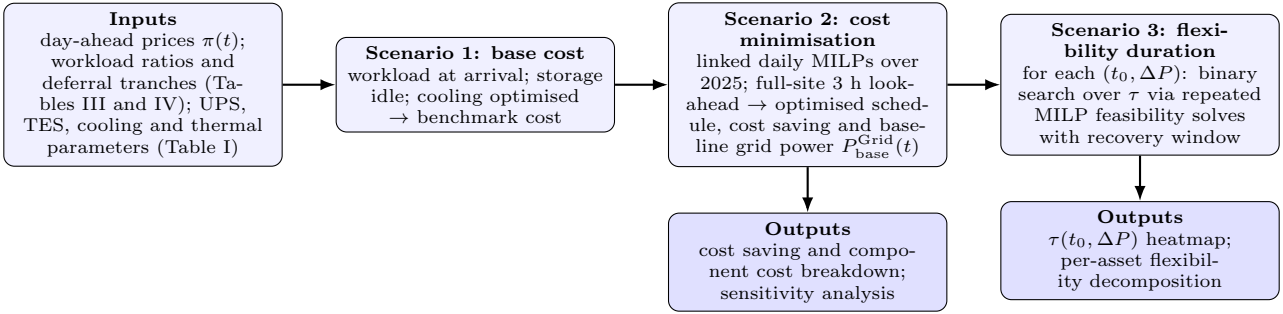

Inputs
day-ahead prices $\pi(t)$; workload ratios and deferral tranches (Tables III and IV); UPS, TES, cooling and thermal parameters (Table I)
Scenario 1: base cost
workload at arrival; storage idle; cooling optimised → benchmark cost
Scenario 2: cost minimisation
linked daily MILPs over 2025; full-site 3 h look-ahead → optimised schedule, cost saving and baseline grid power $P^{\mathrm{Grid}}_{\mathrm{base}}(t)$
Outputs
cost saving and component cost breakdown; sensitivity analysis
Scenario 3: flexibility duration
for each $(t_0, \Delta P)$: binary search over τ via repeated MILP feasibility solves with recovery window
Outputs
$\tau(t_0, \Delta P)$ heatmap; per-asset flexibility decomposition